SURVEYING & LEVELLING · NOTE SL08

Plane Table Methods — Radiation, Intersection and Traversing

Course topics: SWCE1.4 L9

1. Radiation

1. Reconnoitre the area and prepare an index sketch.
2. Select a station O from which every point is visible; set up the table there and mark o on the sheet.
3. Mark the magnetic meridian.
4. Draw rays from o towards stations A, B, C… with the alidade.
5. Measure OA, OB, OC… and plot them to scale along their rays.
6. Join the plotted points to complete the plan.

Use: small, unobstructed areas; mostly combined with other methods.

2. Intersection

1. Prepare an index sketch and select two commanding stations P and Q; PQ is the base line.
2. Set up at P, plot p, mark the meridian and draw rays to Q and to the objects A, B, C…
3. Measure PQ and plot pq to scale.
4. Shift to Q and orient by back sighting on P.
5. Draw rays to the same objects; where the two rays to an object intersect is its position.

- Keep every angle of intersection between 30° and 120°.
- **Use:** inaccessible objects — points across rivers, mountain peaks — and as a supplementary method.

3. Traversing

1. Prepare an index sketch and establish stations A, B, C…
2. Set up at A, plot a and mark the meridian.
3. Sight B, chain AB and plot b.
4. Shift to B, orient by back sighting on A, sight C and plot c.
5. Repeat around the traverse; fill in details by radiation or intersection.

Check: in a closed traverse the closing error shows on the sheet; you can also check by sighting two or more preceding stations.

Method	Distances chained?	Best for
Radiation	Yes — from one station to every point	Small open areas
Intersection	Only the base line	Inaccessible points
Traversing	Yes — each traverse leg	Running a framework through the area, like a compass traverse

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon

e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Points across a river are best located by plane-table

(a) radiation
(b) intersection
(c) traversing
(d) resection by compass

Q2. In the intersection method, only ___ is chained

(a) every ray
(b) the base line
(c) the traverse legs
(d) nothing

Q3. The angle of intersection of rays should lie between

(a) 10° and 45°
(b) 30° and 120°
(c) 60° and 150°
(d) 90° and 180°

Q4. Radiation needs a station from which

(a) only two points are visible
(b) all points are visible
(c) no points are visible
(d) the meridian is unknown

Answers

Q1 — (b) intersection
Q2 — (b) the base line
Q3 — (b) 30° and 120°
Q4 — (b) all points are visible